Assignment 3:

Estimating the Impact and Value of the Information

Total score: 150
This assignment can be completed individually or by a team of a maximum of three members.
Due date: See the course page

Objectives

The main objective of this project is to estimate the **abnormal market reaction** and the **estimated market-value impact** associated with financial news concerning publicly traded companies. The project aims to:

- Quantify the market reaction associated with financial news.
- Measure abnormal stock-price returns associated with news events while distinguishing company-specific reactions from general market movements.
- Estimate the approximate monetary impact of the abnormal market reaction on a company's market capitalization.
- Develop machine-learning models that predict the market reaction and the estimated monetary impact of financial news.
- Evaluate the effectiveness and limitations of NLP and machine-learning methods for this task.

The project focuses on the market reaction associated with news events, rather than claiming that a particular article alone caused the entire observed price movements.

1. Problem Context

Stock prices fluctuate daily because of company-specific developments, sector-wide trends, market movements, investor sentiment, and other unpredictable factors. Investors form expectations about a company's future cash flows, growth prospects, risks, and overall business performance. These expectations are reflected in stock prices, although the market price may differ from any individual investor's estimate of the company's intrinsic value. Ideally, a company's stock price reflects investor's assessment of the intrinsic value of the business.

However, the **intrinsic value** is inherently unknowable with absolute certainty because it is an estimate of a company's future value and the future is unpredictable.

Financial news can provide information that changes investor's expectations. For example, an unexpected earnings announcement, management change, product announcement, analyst upgrade, or regulatory development may be followed by an unusual movement in a company's stock price. However, simply measuring the percentage change in a stock price does not tell us how much of the movement was associated with the news since the stock may have moved because the overall market moved at the same time.

Therefore, this project uses an event-study methodology to estimate the portion of a stock's return that is not explained by general market movements, called the abnormal return. This abnormal return is then used to estimate:

- The Information Impact Score (IIS) describes the direction and unusualness of the market reaction, and
- The Monetary Information Value (MIV) estimates the approximate abnormal change in the company's market capitalization associated with the news events.

The project then uses NLP and machine learning to investigate whether the textual content of financial news can predict these market reactions.

2. Overall Project Workflow

(1) Financial news and market data collection

Financial news data will be provided. You will collect historical stock-market data for the companies in the **S&P 500** and the S&P 500 market index.

(2) Financial event analysis and training data

(a) Use an event-study methodology to measure the market reaction associated with each company day news event. The event study procedure will:

(b) Calculate daily stock returns.

(c) Estimate the expected return of each stock using a market model.

(d) Calculate abnormal returns.

(e) Calculate a three-trading day **Cumulative Abnormal Return** (**CAR**).

(f) Calculate the unusualness of the CAR using a Z-score.

(g) Convert the result into two target variables for machine learning models:

- **Information Impact Score (IIS)**: a discrete score in the range [–3, +3] representing the direction and unusualness of the market reaction
- **Monetary Information Value (MIV)**: an estimate of the abnormal change in the company's market capitalization, expressed in dollars.

Table 1: Example market reaction

News article	*What happened afterward*	*3-day CAR*	*IIS*
Article A	Stock rose 3%	+3.0%	+2
Article B	Stock fell 3%	–3.0%	–2
Article C	Stock rose 0.3%	+0.3%	0

Table 1 illustrates examples of estimated CAR and IIS for each news for a company. The exact IIS is determined from the Z-score thresholds defined later in this assignment. Therefore, the percentage return and the IIS are related but are not the same measure.

(3) Natural Language Processing and Machine Learning

For the news data, apply NLP methods for text processing and vectorization discussed in class. The market-based variables calculated from the historical market data serve as the target/dependent variables. For example, given a news headline such as "Company reports unexpectedly strong earnings…" a regression model might predict a positive MIV, while a classification model might predict a positive IIS. The goal is therefore to investigate whether the textual content of financial news contains information that can be used to predict the subsequent abnormal market reaction.

3. Datasets and File Organization

S&P 500 companies

To keep the project manageable, the data is restricted to companies in the **current S&P 500 constituent list** as the company universe for the entire **2009 – 2020 study period** because S&P 500 membership changes over time. This simplification introduces survivorship bias, which should be acknowledged as a limitation in the final report. You can obtain the current S&P 500 constituent list from various sources such as Dow Jones, Nasdaq websites, or Wikipedia.

Create a file named "**sp500.csv**" with two columns: **ticker symbol** (or ticker) and **company name**. A ticker symbol is a short and unique code used to identify a publicly traded company or security on a stock exchange such as AAPL, GOOG, NVDA).

Financial news data

Two financial news datasets collected during 2009-2020 by publishers are provided:

- "analyst_ratings.csv" – Headlines collected by Publisher A, which may include the price targets estimated by professional analysts through complex modeling to estimate the current intrinsic value of the company.
- "headlines.csv" – Headlines collected by Publisher B, without price target information.

Both datasets follow the same schema but contain different content:

analyst_ratings*(id, headline, URL, publisher, date, symbol*) and ***headlines****(id, headline, URL, publisher, date, symbol)*

- id: row number
- headline: headline news that may include the price target.
- URL: the URL of the headline news
- Publisher: news publisher
- Date: published date and time
- symbol: ticker representing the company

Create a new file named "**news.csv**" by merging the two datasets. Use the following schema:

news*(date, symbol, headline, URL, publisher, article)*

The article is an optional column (not required in this project) for a full article about the headline you can download it directly from the publisher website if allowed.

If multiple news items are associated with the same company and the same calendar date, combine them into one company-day news event by concatenating the relevant headline/text fields with a separator. The resulting unit of analysis is therefore a company-day news event, rather than an individual article. This distinction is important when multiple news items occur for the same company on the same day, the subsequent market reaction cannot generally be attributed to one particular article.

Financial price data

Download historical daily prices for every ticker in "sp500.csv" and the S&P 500 index using the ticker name "^sp500" (^GSPC in yahoo finance) for the same period as the news data. You can freely download the data from sources such as Yahoo Finance using any available Python package (e.g., "yfinance" or "yahoo-fin"). Save the data **locally** in a file named "historical_prices.csv".

For each company, the required fields are "date", "symbol", "open", "high", "low", "close", "adj_close", "volume", and "market_cap". For consistency, treat the daily S&P 500 index values like stock prices with the symbol "^sp500", and include them in historical_prices.csv. The schema for historical_prices.csv will be:

historical_prices*(date, symbol, open, high, low, close, adj_close, volume, market_cap)*

For "market_cap", if historical market capitalization is available directly from the data source, use that value. If it is not available, estimate it using an appropriate historical share price and historical shares outstanding, for example:

- $marke_{cap} \approx number\ of\ outstanding\ shares\ \times close\ price$ or
- $marke_{cap} \approx market_cap_{current} \times (\frac{close_t}{close_{current}})$.

If the market capitalization is calculated, use "close" price instead of "adj_close" price. Clearly document the method and assumptions used to estimate the market capitalization.

For the S&P 500 index row with the symbol "^sp500", "market_cap" should be left blank/NaN because the S&P 500 index is not an individual company.

Organization of Program Files and Directories

The news datasets, required program files, sample files, and directory structure are provided in "Assignment_3.zip" for efficient grading. Before getting started, review the files. These files are provided to help standardize and validate your CSV files. Most files in the directories are currently blank and must be completed by you. **Do not change** any file names, directory names, or the directory structure.

4. Stock Returns, Volatility, and Abnormal Returns

Daily returns

Daily return is a method of calculating the percentage change in a stock price from one trading day to the next as the simple daily return $R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} = \frac{P_{i,t}}{P_{i,t-1}} - 1$ where $P_t = adj_close_t$ (adj_close price) at time t and $P_{t-1} = adj_close_{t-1}$ at previous time.

Daily volatility

Daily volatility is the standard deviation of daily returns over a predefined historical window. For this project, use a **60-day-trading-day rolling window**. For example, the 60-day rolling daily volatility is: $\sigma_{i,t} = SD(R_{i,t-59}, R_{i,t-58}, \dots, R_{i,t})$ where $R_{i,t}$ = daily return of company i on day t.

Volatility measures the variability of returns and is commonly used as a measure of market risk. If volatility for other intervals is required, it can be calculated: $\sigma_{period} = \sigma_i \times \sqrt{time}$. Based on this formula, assuming returns are uncorrelated and have stable variance, annualized volatility approximated by: $\sigma_{annual} = \sigma_{daily} \times \sqrt{252}$ where 252 = the approximate number of trading days in a year and σ_{daily} = the daily volatility. In simple terms: if daily volatility is 2%, annualized volatility is approximately 2% $\times \sqrt{252} \approx 31.7\%$. Volatility can be interpreted as risk. Risks accumulate additively in variance, so annual volatility can be considered as yearly risk.

Market model

A market model in finance is a statistical technique that breaks down a specific security's return into a portion driven by the overall market and a portion unique to the company. Stock prices are influenced by both company-specific events and broader market movements (typically represented by major indices such as Dow Jones, S&P 500, NASDAQ). To isolate the company-specific component of stock returns, we will use a market model based on the S&P 500 index. A market model can be defined: $\boldsymbol{R_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t}}$

where:

- $R_{i,t}$ = return of stock i at time t

- $R_{m,t}$ = market (S&P 500) return at time t
- α_i = intercept
- β_i = sensitivity to the market
- $\epsilon_{i,t}$ = residual

The market model separates the portion of a stock's return associated with general market movements from the portion not explained by the market model.

Alpha (α) is the estimated intercept of the market model and represents the expected return when the market return is zero, according to the fitted model. So, $\alpha > 0$, stock outperforms what's expected given its beta; $\alpha < 0$, stock underperforms what's expected given its beta.

Beta (β) measures how sensitive a stock is to the market. So, $\beta = 1$, stock moves in line with the market; $\beta > 1$, the stock is more volatile than the market; $\beta < 1$, stock is less volatile than the market; $\beta < 0$, the stock moves opposite to the market. However, β should not be interpreted as the stock's total volatility.

Residual (ϵ_t) is the portion of the stock return not explained by the market. It may reflect company-specific information, industry effects, other omitted factors, or random variation.

Estimation window

Estimate α and β using historical returns before the news event. For this project, use **120 trading days** (for estimating α and β) **ending 20 trading days** (gap) **before the event date**. Thus, the estimation window is separated from the event window. Do not use observations from the event window to estimate the market-model parameters. This prevents information from the event period from contaminating the expected-return model.

Event date and news timestamp

The event date is the first trading day on which the market could reasonably react to the news. For simplicity, use the following rules:

- News published during regular market hours is assigned to that trading day.
- News published before market open is assigned to that trading day.

- News published at or after 4:00 PM Eastern Time is assigned to the next trading day.
- News published on a Saturday, Sunday, or holiday (when the market closed) is assigned to the next available trading day.

The publication timestamp should be interpreted using the appropriate time zone before applying these rules.

Expected returns and abnormal returns

Using the market model, the **expected return** of a stock is: $E(\hat{R}_{i,t}) = \hat{\alpha}_i + \hat{\beta}_i R_{m,t}$.

The **model parameters $\hat{\alpha}$** and **$\hat{\beta}$ are estimated values for α** and **β** by regressing historical daily returns preceding the news event to **avoid** look-ahead bias. The **Abnormal Return** (**AR**) is calculated as: $\boldsymbol{AR_{i,t} = R_{i,t} - E(\hat{R}_{i,t}) = R_{i,t} - \hat{\alpha}_i + \hat{\beta}_i R_{m,t}}$.

$AR_{i,t}$ is the portion of stock returns not explained by general market movements. You can estimate $\hat{\alpha}$ and $\hat{\beta}$ from $R_{i,t} = \alpha_i + \beta_i R_{m,t}$ through regression (e.g., the Ordinary Least-Squares) over the estimation window (120-day with 20-day gap before each news event).

Cumulative Abnormal Returns (CAR)

For this project, **assume** that the market reaction to a news event is measured over a three-day event window consisting of the news day and the following two trading days, then abnormal return for a three-day news window: $\boldsymbol{CAR_{t,t+2} = AR_{i,t} + AR_{i,t+1} + AR_{i,t+2}}$. Therefore, CAR is the sum of abnormal simple returns. For example,

Day	*Stock*	*S&P500*	*Expected*	*Abnormal*
News day	+5.0%	+1.0%	+0.8%	+4.2%
Day +1	+1.0%	+0.2%	+0.2%	+0.8%
Day +2	-0.5%	-0.1%	-0.1%	-0.4%

Over the three-day event window, the stock generated 4.6% points of cumulative abnormal performance relative to what the market model predicted. That's much more informative for a news impact analysis than simply saying "The stock increased by 5.5%," since the 5.5% may include returns that would have occurred anyway due to normal market movements and the stock's sensitivity to the market.

The Monetary Information Value (MIV)

CAR measures the abnormal percentage change. The monetary information value estimates the approximate abnormal change in the company's market capitalization associated with the news event, can be estimated: $MIV_{i,t} = MC_{pre} \times CAR_{i,t}$

where:

MC_{pre} is the company's market capitalization immediately before the news event.

$CAR_{i,t}$ is the three-day cumulative abnormal return.

For example, if a company has a pre-news market capitalization of $20 billion and a three-day CAR of +2.5%: MIV = $20B x 0.025 = $500M

The estimated monetary information value $500M represents the abnormal increase in market capitalization associated with the news over the event window. A negative value indicates an abnormal decrease in market capitalization. Note that this is an estimated market-value impact, not proof that the article alone caused the entire change.

Z-score as a measure of the unusualness of the market reaction

A large CAR does not necessarily represent unusual information since a volatile stock can experience large price movements even without significant news information. Therefore, standardize the CAR using the historical distribution of three-day CARs. For each company, calculate historical three-day CARs during the estimation period.

Let:

CAR_{event} = CAR for the current news event

μ_{CAR} = mean of the company's historical three-day CARs

σ_{CAR} = standard deviation of the company's historical three-day CARs

The Z-score is: $Z_{CAR} = \frac{CAR_{event} - \mu_{CAR}}{\sigma_{CAR}}$. For a news event occurring on trading day t:, calculate the historical CARs as follows:

(1) Estimate $\hat{\alpha}$ and $\hat{\beta}$ using the 120 trading days ending 20 trading days before t.

(2) Use those same $\hat{\alpha}$ and $\hat{\beta}$ to calculate historical abnormal returns.

(3) Calculate the historical three-day CARs, CAR_{event} within that 120-trading-day estimation window.

(4) Calculate the mean, μ_{CAR} and standard deviation of those historical CARs, σ_{CAR}.

(5) Use them to standardize the event CAR, Z_{CAR}.

The Z-score tells us how unusual the reaction was relative to the company's historical behavior. If σ_{CAR} is zero or too small to produce meaningful Z-score, the event should be excluded and the reason documented.

Information Impact Score (IIS)

Map the Z-score into a discrete score from –3 to +3:

Z-score	*Information Impact Score (IIS)*	*Interpretation*
$Z \leq -3$	–3	Extremely negative
$-3 < Z \leq -2$	–2	Strongly negative
$-2 < Z \leq -1$	–1	Moderate negative
$-1 < Z < 1$	0	Neutral, normal
$1 \leq Z < 2$	+1	Moderate positive
$2 \leq Z < 3$	+2	Strong positive
$Z \geq 3$	+3	Extremely positive

The IIS represents both the direction of the abnormal market reaction and its unusualness relative to the company's historical behavior.

Data preparation for modeling

Complete the program "**1_calculate_impacts_values.py**". This program should create "historical_impacts_values.csv" based on "historical_prices.csv". The schema of the dataset should be:

historical_impacts_values*(date, symbol, open, high, low, close, adj_close, volume, market_cap daily_return, daily_volatility, market_return, alpha, beta, AR, CAR, Z_CAR, IIS, MIV)*

where CAR represents the three-trading-day cumulative abnormal return beginning on that date.

The program "1_calculate_impacts_values.py" should be executable with the command:

"**python** 1_calculate_impacts_values.py" without requiring additional parameters or instructions.

5. Text Preprocessing and Sentiment Terms

From the news datasets, **identify 5 sentiment-bearing or financially informative words/terms** that may be associated with positive or negative stock-market reactions, e.g., gain, loss, strong, weak, upgrade, downgrade, etc.

Briefly justify your choices using evidence from the dataset and/or relevant financial news or sentiment concepts discussed in class.
Remember that individual words do not necessarily determine sentiment independently of context. For example the word "loss" may occur in a statement describing an improvement in losses.

The headlines or full news articles may be provided in plain text or HTML format and can contain unnecessary text content. As discussed in class, the possible text preprocessing tasks are:

- **Tokenization and text normalization**: extracting tokens, lowercasing, removing punctuation, etc.
- **Relevant text extraction**: to ensure that only the main content of the news article or headline is retained, removing irrelevant contents such as HTML tags, advertisement, navigation menus, or unrelated HTML content.
- **Additional text processing**: to handle stop words, negation, occurrence of important words, etc.

Briefly describe the preprocessing methods you applied and justify the importance of the method.

6. Training and Testing Datasets

Data sampling

Complete the program "2_sampling_vectorize.py". The program should randomly select a consecutive 7-year period from the available study period using a np.andom.seed(42) (with

possible start years [2009, 2010, 2011, 2012, 2013, 2014]) to ensure reproducibility, then split the selected period chronologically into **80% training** and **20% testing**.

The selected segment must preserve chronological order to avoid mixing future data into the past and to **prevent lookahead bias**. Clearly specify **beginning date**, **ending date**, and **sampling method**. To prevent **target leakage caused by the three-trading-day event window**, an observation is included in a dataset only if all three trading days used to calculate its CAR fall within the corresponding training or testing period. Therefore, training observations whose event windows extend into the testing period are excluded.

Training and testing data

After selecting the seven-year period, match the news events with their corresponding market-based target variables from "historical_value_impact.csv", and create:

news_dtm_train*(date, symbol, news_vector, IIS, MIV)*

news_dtm_test*(date, symbol, news_vector, IIS, MIV)*

news_tfidf_train*(date, symbol, news_vector, IIS, MIV)*

news_tfidf_test*(date, symbol, news_vector, IIS, MIV)*

Important vectorization procedure:

To prevent data leakage, the vectorizer must be fit using the training corpus only. The procedure should be:

(1) Fit the vectorizer using the training document

(2) Transform the training documents using the fitted vectorizer

(3) Transform the testing documents using the same fitted vectorizer

Do not independently fit the vectorizer on the testing corpus. So, for TF-IDF, this means that:

- The vocabulary is determined from the training data
- IDF values are calculated from the training data
- The same vocabulary and IDF values are used to transform the test data

This ensures that the test data remains genuinely out-of-sample. The same principle should be followed for DTM.

7. Machine Learning Modeling

Information Impact Score Classification

Complete the program "**3_classify.py**" for classification modeling.

Design an MLP in PyTorch, clearly describing the key architectural choices:

- Number of layers, Number of neurons per layer, Activation functions
- Other hyperparameters (e.g., learning rate, batch size, optimizer, number of epochs).

Train a classifier on **each vectorized dataset** (DTM and TF-IDF) and evaluate it using the test dataset in terms of % accuracy. **Compare** the model performance between **DTM** and **TF-IDF** vectorization under the same modeling and evaluation conditions.

Report the results for the top 10 companies with the highest number of test observations.

Discuss the results of the classification modeling in the report.

Monetary Information Value Regression

Complete the program "**4_regression.py**".

Design an MLP regression model using PyTorch, clearly describing the key architectural choices:

- Number of layers, Number of neurons per layer, Activation functions
- Other hyperparameters (e.g., learning rate, batch size, optimizer, number of epochs).

Train a regression model on **each vectorized dataset** and evaluate it using the test dataset.

Report regression performance for the top 10 companies selected according to the criteria Root Mean Square Error (RMSE) and R^2. **Compare** the model performance between **DTM** and **TF-IDF** vectorization under the same modeling and evaluation conditions.

Report the results for the **top 10 companies** with the highest number of test observations.

Discuss the results of regression modeling in the report.

Analysis and Reflection

Briefly **summarize** your overall results, highlighting key observations. Your report should evaluate the effectiveness of the methods, any limitations, data constraints, challenges faced, insights gained, and lessons learned from this project.

Suggest any potential improvements or alternative approaches that could achieve better results in future studies.

8. Deliverables and Grading Criteria

One report file (PDF or Word) that includes:

- All team member names and % contribution of each member. If every member contributed equally, simply state "equal contribution." If there is disagreement on contributions, provide a brief task description for each member. In this case, different grades may be assigned individually.
- All answers and results from the required tasks
- When including example data in the report, show only a few representative samples instead of the entire dataset.

All Python program files you created and only **CSV files generated** from your programs. Do not upload the original news datasets or Docker images.

Grading Criteria

- Quality of the work shown in the report (e.g., modeling process, implementation, results, correctness, and analysis)
- Level of understanding of related subjects as reflected in the report
- Effort (10%) demonstrated in the project